NeST-Com: Fixed-Budget Next-Scale Token Communication

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Abstract

NeST-Com is a digital image communication framework that uses next-scale residual descriptions under fixed per-image symbol budgets. It coordinates source-prefix selection, channel protection and prefix-constrained reconstruction through actual-length frame fitting. Independently terminated within-scale prefixes retain scale-wise probability evaluation, while received boundary tokens refine the residual state and finer-scale predictions. At 1024 symbols and 10 dB, source-bit utilization increases from 77.70% to 99.79%, reducing DISTS and DreamSim by 8.7% and 20.5% relative to shared-model whole-scale transmission. Matched controls and complete-system comparisons characterize reconstruction gains and quality–cost tradeoffs.

Index Terms

Image communication, next-scale tokens, fixed symbol budget, perceptual reconstruction.

I. Introduction

A fixed per-image digital frame must accommodate source information, signaling and protection. The transmitted description should use the available capacity to anchor reconstruction, while a learned prior predicts untransmitted content.

Next-scale representations provide complete coarse maps followed by residual refinements [1]. Each coarse map covers the full latent field and supplies context for the next map's probabilities. ARPC develops this structure into progressive compression through scale selection, entropy coding, generation and adaptive quantization [2]. Ada-TokenCom combines next-token prefix coding and tail generation with prefix-length/MCS adaptation under a long-term symbol budget [3]; ResARC codes suffix-generation residuals [4].

We propose NeST-Com, a digital communication framework that connects next-scale source descriptions, channel protection and prefix-constrained reconstruction under a fixed per-image symbol budget. Its central design problem is matching image-dependent source lengths to protected-frame capacity: a whole residual map may not fit even when many of its tokens would. This motivates finer transmission endpoints within the progressive description [5].

The framework retains scale-wise probability evaluation and fits independently terminated within-scale prefixes by their actual lengths. The received prefix constrains residual reconstruction and finer-scale prediction through VAR's known-token overwrite and recursive update [1]. A coder-specific length property characterizes feasible prefixes. Shared-model policies and matched controls establish the fitting gains; receiver interventions explain information use, and complete-system comparisons quantify quality–cost tradeoffs.

II. Proposed NeST-Com Framework

A. Next-Scale Representation and Digital Frame

We transmit $x \in [0,1]^{256\times256\times3}$ using N complex symbols over AWGN,

$$y = s + n, \quad n \sim \mathcal{CN}(0, \sigma^2 I_N), \quad \mathrm{SNR} = 10\log_{10}(E_s/\sigma^2), \tag{1}$$

with nominal $E_s = 2$. TX uses the frozen policy for its design SNR; RX uses the actual noise variance for LLRs. Digital constellations have average energy two without per-frame

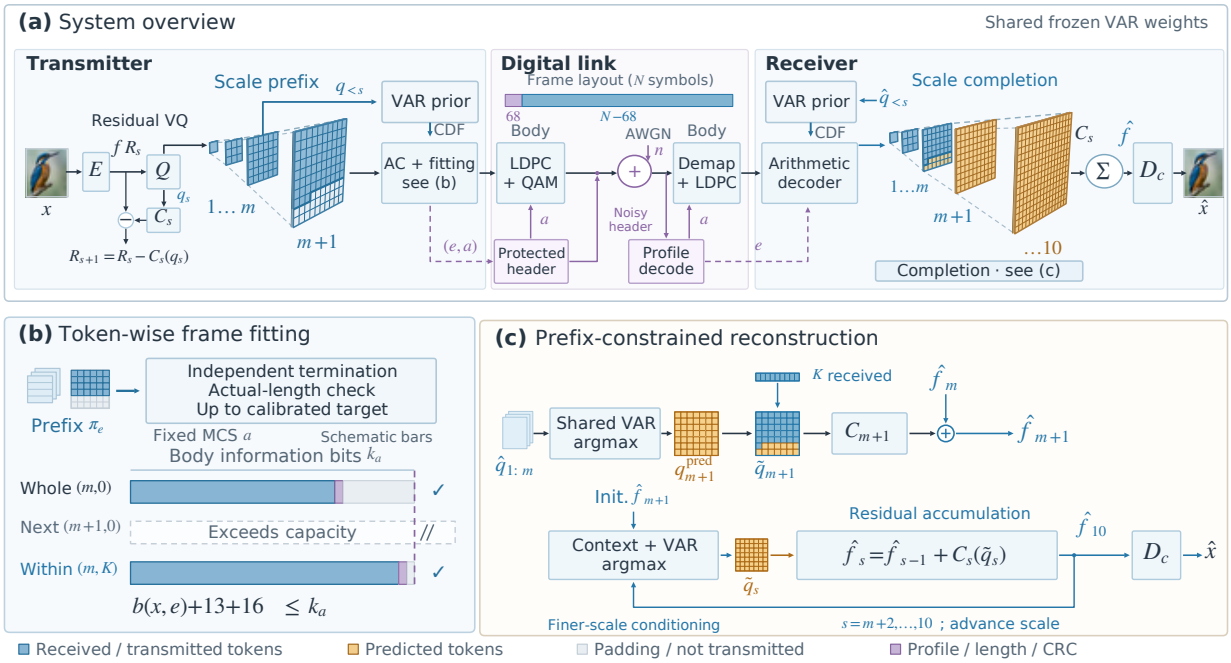


Fig. 1. NeST-Com. (a) The shared policy fixes the MCS and profile directory; the protected profile selects the endpoint. VAR and known coarse tokens provide coding probabilities. (b) Independently terminated prefixes fit the actual capacity below the target; bars are schematic. (c) Received tokens overwrite boundary predictions. Recursive state updates accumulate residuals and condition each finer scale before decoding. Unreceived boundary positions retain their coarse-conditioned predictions.

renormalization. A 12-bit profile ID, CRC16, and terminated rate-1/2 convolutional code occupy 68 rotated-BPSK symbols $\{\pm(1+j)\}$. Under a preconfigured policy, both terminals know the MCS and profile directory; the 12-bit profile identifies the endpoint within that directory. The remaining $N-68$ symbols carry the LDPC-coded body, with public frame-dependent scrambling after rate matching and inverse scrambling before decoding. Continuous JSCC data are normalized per image to energy two; required side information is paid within N.

The frozen VAR tokenizer produces ten residual token maps q_s, with side lengths $(1,2,3,4,5,6,8,10,13,16)$ and a 4096-entry codebook [1]. Write L_s for the number of positions and C_s for codebook lookup, resizing to the full latent grid, and the scale-dependent

residual transform. The latent sum $\sum_s C_s(q_s)$ progressively refines a full-field description. Decoder D_c is adapted on 20 000 ImageNet training images using MSE plus 0.1 LPIPS, then frozen. The continuous/quantized/prefix-completed training mixture is specified in Supplement S1. The null-class VAR-d16 prior factorizes as

$$p_\theta(q_s \mid q_{<s}) = \prod_{i=1}^{L_s} p_\theta(q_{s,i} \mid q_{<s}). \tag{2}$$

Under the specified numerical route (S1), both terminals reconstruct identical integer CDFs, with total frequency 2^{24}, for arithmetic coding [6]; no unreceived same-scale token is needed. Transmission can therefore end within a scale while probability evaluation remains scale-wise.

B. Within-Scale Endpoints and Actual-Length Fitting

An endpoint $e = (m, K)$ retains complete coarse maps and a raster-ordered boundary prefix:

$$\pi_e = (q_1, \ldots, q_m, q_{m+1,1:K}). \tag{3}$$

For $K = 0$, including $e = (10, 0)$, the boundary prefix is empty. Raster order specifies positions without a mask. WHOLE permits $(m, 0)$ for $m = 4, \ldots, 10$. NeST permits every integer $K = 0, \ldots, L_{m+1} - 1$ for $m = 4, \ldots, 9$, together with $(10, 0)$: 651 endpoints from 30 through 680 retained tokens. Quarter/Eighth retain $K = \lfloor jL_{m+1}/d \rfloor$, $j = 1, \ldots, d-1$, for $d = 4/8$, plus complete-scale boundaries, with duplicates removed. Token is the main configuration.

For MCS $a = (M, r)$, the coded body has $n_a = (N - 68)\log_2 M$ bits and information length $k_a = \lfloor r n_a \rfloor$. Its independently terminated arithmetic length $b(x, e)$ must satisfy

$$b(x, e) + 29 \le k_a, \qquad \eta(x, e) = \frac{b(x, e)}{k_a - 29}. \tag{4}$$

The 29 bits comprise a 13-bit length and CRC16. Remaining positions are zero-padded before 5G NR LDPC coding, rate matching [7], and QAM; η measures source-bit capacity utilization after these 29 overhead bits.

For $N = 1024$, WHOLE, NeST and Ada use the same 12 MCSs: QPSK/16-QAM/64-QAM with rates $1/2, 2/3, 3/4, 5/6$. WHOLE has seven ceilings; NeST uses 25 Quarter ceilings with Token fallback; Ada allows prefixes through 256. All 396 candidates enter a 100-image, one-realization pilot. Per method/SNR, the top three plus a preset QPSK rate-1/2 anchor advance to 1000 sources with three realizations. Maximum equal-source DINOv2 cosine [8], including gray failures, selects the policy. $N = 512$ retains its independently calibrated policies (Table IV).

A public method/MCS directory, frozen before calibration and shared across images, maps each profile to an endpoint. One scale-wise probability and incremental coding traversal supplies the candidates. Non-mutating termination makes each stream independently decodable; streams are not slices of a longer codeword.

For this coder's fixed token/CDF traversal and non-mutating finish, $b_t = E_t + P_t + 2$ after t tokens, where E_t and P_t count emitted and pending bits. Renormalization increments $E_t + P_t$; other interval updates preserve it. Thus b_t is nondecreasing, and feasible endpoints form an initial segment for fixed MCS and ceiling. TX selects its longest prefix, or sends a paid NULL profile with a two-bit zero payload if none fits. The measured code scans all candidates; S1 gives algorithm and proof details.

C. Prefix-Constrained Reconstruction

Invalid profile CRC/catalogue, body CRC, declared length, known padding, or arithmetic parsing yields a fixed gray image, as does valid NULL reception. Otherwise, RX decodes the actual prefix using the shared CDFs. At $e = (10, 0)$, all maps are received and directly decoded. For an incomplete prefix, RX applies VAR's known-token overwrite and recursive state update [1]. It predicts the boundary map by argmax conditioned on complete received

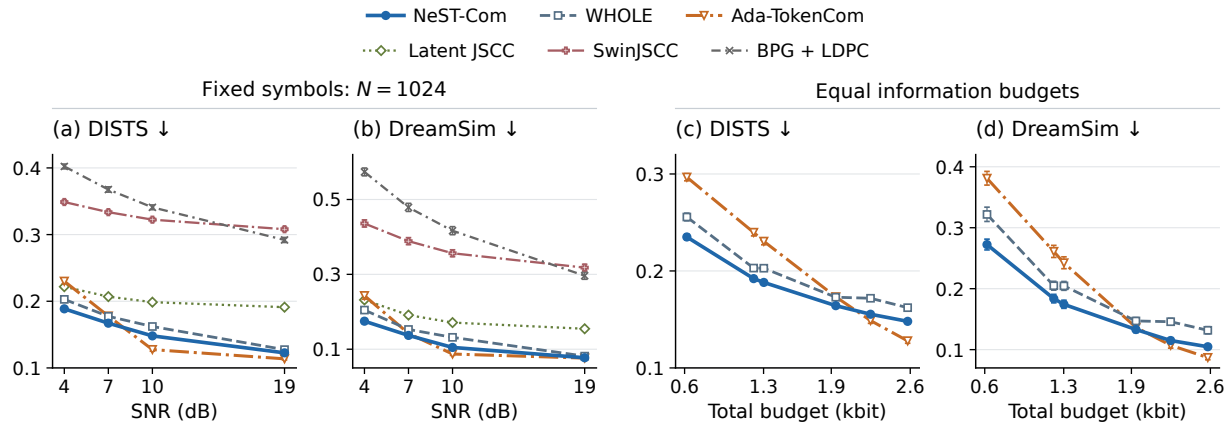


Fig. 2. Source-referenced reconstruction on 500 ImageNet images. (a,b) Six methods at $N = 1024$; WHOLE, NeST and Ada share the 12-MCS calibration procedure. Three outcomes per source include gray failures. (c,d) Equal, error-free information budgets pay 57 shared protocol bits and Ada's additional 10-bit class; WHOLE/NeST share ceiling $(10, 0)$. Points and bars are equal-source means and pointwise 95% source-bootstrap intervals after within-source averaging. Ada-TokenCom is the predicted-class, fixed-frame adaptation. Lower values are favorable.

maps, then overwrites received positions:

$$\tilde{q}_{m+1,i} = \begin{cases} \hat{q}_{m+1,i}, & i \le K, \\ q^{\text{pred}}_{m+1,i}, & i > K. \end{cases} \tag{5}$$

Received boundary tokens serve two roles: they correct the residual representation and, through the merged boundary map, condition finer-scale prediction. Unreceived boundary positions retain their original coarse-conditioned prediction; there is no same-scale prediction loop. Complete received maps remain unchanged. After generating the finer maps,

$$\hat{x} = D_c\left(\sum_{s=1}^{10} C_s(\tilde{q}_s)\right). \tag{6}$$

At the last scale, overwrite only changes the residual contribution.

III. Experimental Results

A. Experimental Setup

Common500 ImageNet validation sources [9] are disjoint from cal1000, with fixed 256×256 preprocessing. We test $N = 1024$ at 4/7/10/19 dB and $N = 512$ at 4/10/19 dB. Sources first average three realizations including gray failures; equal-source means and direct paired 95% intervals use 10 000 source resamples. Controls use control100. Calibrated token systems and matched controls share noise; external systems retain historical streams.

Latent JSCC maps tokenizer latents to symbols and shares D_c (MSE, LPIPS and latent-consistency losses, 100k updates). SwinJSCC [10] uses RGB MSE (200k updates, calibration-MSE selection). Both train on 20 000 images at $N = 1024$. BPG [11] uses LDPC/QAM. Ada-TokenCom [3] is a predicted-class, fixed-frame adaptation: LlamaGen-L/VQ16, prefix arithmetic coding, CFG=4, $T = 1$, top-$k = 2000$, top-$p = 1$, and a paid 10-bit ConvNeXt class. Long-term control and ARQ are excluded. Supplement S1 specifies training, numerical routes and our ARPC digital adaptation.

DISTS [12], DreamSim [13], PSNR, LPIPS-Alex [14] and CLIP-I [15] use native float RGB and fixed preprocessing (S1). Timing uses 32 sources, two warmups, three measurements, batch one, one RTX PRO 6000 and two physical CPU cores. Synchronized RGB-to-symbols and symbols-to-RGB spans include coding, link processing and generation, excluding loading, I/O and scoring (S7).

B. Within-Scale Fitting and Granularity

Table I(A) compares NeST and WHOLE with shared model, coder and receiver under frozen family policies. All five paired metrics favor NeST across seven budget/SNR settings. At $N = 1024$, 10 dB, source-bit capacity utilization rises from 77.70% to 99.79%, DISTS falls from

TABLE I

WITHIN-SCALE TRANSMISSION. (A) SHARED-MODEL NEST–WHOLE DIFFERENCES. (B) MATCHED ENDPOINT GRANULARITY. (C) ORDERED RECEIVER INTERVENTIONS. BRACKETS ARE SOURCE-PAIRED 95% INTERVALS; SAMPLE SIZES ARE SPECIFIED FOR EACH BLOCK.

A. 500 sources, $N = 1024$, uniformly calibrated policies

SNR	Util. W→N (%)	ΔDISTS	ΔDreamSim
4	73.81→99.56	-0.0142 [-0.0155, -0.0127]	-0.0297 [-0.0330, -0.0264]
7	83.19→99.71	-0.0103 [-0.0139, -0.0068]	-0.0159 [-0.0207, -0.0112]
10	77.70→99.79	-0.0141 [-0.0153, -0.0128]	-0.0270 [-0.0299, -0.0241]
19	85.24→99.83	-0.0048 [-0.0053, -0.0043]	-0.0043 [-0.0051, -0.0036]

B. 100 quality sources; 32 timing sources; ceiling $(8, 126)$

Grid	Util. (%)	DISTS	DreamSim	TX (ms)
Whole	78.13	0.1650	0.1387	185.2
Quarter	95.19	0.1535	0.1143	215.8
Eighth	97.44	0.1516	0.1137	217.8
Token	99.85	0.1505	0.1115	248.4

C. 100 sources, 10 dB, same received prefixes

Intervention	ΔPSNR (dB)	ΔDISTS	ΔDreamSim
B0–A0	+0.287 [0.240, 0.335]	-0.0062 [-0.0077, -0.0048]	-0.0113 [-0.0147, -0.0083]
C0–B0	+0.537 [0.404, 0.682]	-0.0083 [-0.0102, -0.0065]	-0.0159 [-0.0205, -0.0116]

A0: prior completion without received boundary tokens. B0: A0 plus boundary residual correction. C0: B0 plus updated finer-scale conditioning.

0.1621 to 0.1481, and DreamSim from 0.1317 to 0.1048. PSNR gains span 0.49–0.88 dB across the four $N = 1024$ points.

To isolate endpoint granularity, Block B fixes model, 16-QAM rate 2/3, ceiling $(8, 126)$, fitting rule and noise. Within-scale grids convert unused capacity into better reconstruction. Eighth retains approximately 92% of the Whole-to-Token gain in PSNR, DISTS and DreamSim. Token raises utilization from 97.44% to 99.85% and adds $0.066\,[0.034, 0.101]$ dB PSNR, with DISTS/DreamSim differences $-0.0011\,[-0.0020, -0.0002]$/$-0.0022\,[-0.0042, -0.0001]$, at 30.7 ms (14.1%) additional TX processing. Eighth offers a lower-cost operating point; Token prioritizes fuller capacity use and the additional measured reconstruction gain.

The $N = 512$/10-dB control fixes 100 sources, three noise realizations, model, receiver, 16-QAM rate 2/3 and ceiling $(7, 0)$. Token–Whole improves PSNR by $0.534\,[0.446, 0.625]$ dB; Eighth–Whole gives $0.464\,[0.383, 0.547]$ dB, retaining about 87% of the mean gain. Token–Eighth adds $0.070\,[0.029, 0.115]$ dB and improves LPIPS/DreamSim, while DISTS/CLIP-I intervals cross zero (S3). Finer termination thus remains useful at the reduced budget.

Block C explains how received boundary information improves reconstruction through ordered interventions on the same actual NeST prefixes. A0 completes boundary and finer maps without received boundary tokens. B0 adds their residual correction while retaining A0's finer-scale conditioning; C0 also updates that conditioning. At 10 dB, B0–A0 improves PSNR by 0.287 dB and reduces DreamSim by 0.0113; C0–B0 adds 0.537 dB and a 0.0159 reduction. Paired intervals favor both stages for these metrics and DISTS. Supplement S3 includes 7 dB and the final-scale case, where no finer-scale propagation remains.



Fig. 3. Selected reconstructions at $N = 1024$ and 10 dB. WHOLE, NeST and the predicted-class fixed-frame Ada-TokenCom adaptation use the policies in Table IV. Values are full-image DISTS↓ / DreamSim↓ for each method's first prescribed realization. Supplement S6 gives the fixed source identities, selection details and additional examples.

TABLE II

AUXILIARY QUALITY ON COMMON500 AT $N = 1024$, 10 DB. MEANS INCLUDE THREE REALIZATIONS PER SOURCE AND GRAY FAILURES. POLICY-SPECIFIC TIMING IS REPORTED SEPARATELY IN S7.

Method	PSNR (dB)↑	LPIPS↓	CLIP-I↑
WHOLE	19.31	0.199	0.858
NeST-Com	20.19	0.166	0.877
Ada-TokenCom	19.56	0.167	0.884
Latent JSCC	20.80	0.173	0.789
SwinJSCC	23.03	0.469	0.669
BPG + LDPC	22.54	0.589	0.635

C. Perceptual Reconstruction and Processing Cost

Fig. 2(a,b) compares six methods at equal symbol budgets, with WHOLE as the shared-model reference. External comparisons include representation, coding and receiver differences. NeST has lower DISTS/DreamSim than SwinJSCC, Latent JSCC and BPG at all four $N = 1024$ SNRs (24 paired intervals excluding zero). SwinJSCC and BPG have higher PSNR, showing a pixel-fidelity/perceptual-similarity tradeoff.

Against Ada, NeST improves both distances at 4 dB and DISTS at 7 dB; at 7 dB, the DreamSim difference interval crosses zero. Ada yields lower distances at 10 dB and lower DISTS at 19 dB, where DreamSim again crosses zero. NeST has higher PSNR throughout. At 7 dB, WHOLE/NeST/Ada reject 11/7/10 of 1500 receptions each; NeST rejects one at 4 dB, with zero rejections at the other token-system points and no NULL outputs. Table II and S2 give auxiliary metrics. NeST also has lower DISTS/DreamSim than our specified ARPC digital-link adaptation at all four SNRs (S2).

Fig. 2(c,d) isolates source-rate behavior at six equal information budgets, 620–2577 bits, without channel errors. Totals pay 57 shared overhead bits plus Ada's 10-bit class. WHOLE/NeST share ceiling $(10, 0)$; Ada permits 256 tokens and averages three generation repeats per source. NeST improves all five metrics over WHOLE throughout. Against Ada, DISTS favors NeST at the four smallest budgets and DreamSim at the first three; the 1940-bit DreamSim interval crosses zero. Ada has lower distances at 2248/2577 bits.

On 100 fixed sources under the earlier policies, changing only Ada's tail sampling to argmax increases DreamSim by $0.0450\,[0.0282, 0.0621]$, $0.0345\,[0.0232, 0.0461]$ and $0.00358\,[0.00101, 0.00601]$ at N/SNR=512/4, 512/10 and 1024/10 (S3). Thus completion choice contributes to the observed quality tradeoff.

Fig. 3 illustrates NeST preserving truck markings and foreground contours. These selected cases favor NeST in DISTS/DreamSim and SwinJSCC in PSNR; S6 gives identities, selection and broader examples.

In the measured H10 configuration, NeST's TX/RX costs are 248.44/403.99 ms, versus 215.29/366.42 ms for WHOLE; totals are 652.43 ms for NeST and 39.17 ms for SwinJSCC (S7). H10 shares U10's MCS and ceilings but uses earlier directories (S1.3). Compiled Ada's 3.064-s total is paired with its own 32-source quality (S7.4–S7.5).

D. Reduced Budget and Transfer to Kodak

At $N = 512$, all five NeST–WHOLE intervals favor NeST at each SNR, with PSNR gains of 0.50–0.99 dB. At 10 dB both use target $(7, 0)$ and 16-QAM rate 2/3; utilization rises from 79.30% to 99.54%. Against Ada, NeST has higher PSNR and lower LPIPS throughout, and lower DISTS/DreamSim at 4/10 dB. Ada is better on both distances at 19 dB: NeST–Ada differences are $0.0058\,[0.0036, 0.0079]$ and $0.0072\,[0.0037, 0.0106]$. Supplement S2 gives CLIP-I and rejection counts.

With the earlier ImageNet-calibrated policies, Kodak24 center crops give Token–WHOLE PSNR gains of 0.567/0.794/0.405 dB at 4/10/19 dB and favorable DISTS/DreamSim intervals throughout. Against Ada, Token has lower distances at 4 dB; Ada has lower DISTS at 10/19 dB, where DreamSim intervals cross zero. All 864 events succeed (S8). Historical Ada at 19 dB uses 64-QAM rate 5/6; its current policy uses 16-QAM rate 5/6 (Table IV).

E. Operation Near the Channel Threshold

Fixing H10 policies, 100 sources and 20 noise realizations yield 30 000 receptions over 8–10 dB. All methods reject every frame at 8 dB. WHOLE/NeST/Ada rejection rates are

TABLE III

REDUCED-BUDGET RECONSTRUCTION ON 500 SOURCES AT $N = 512$. MEANS INCLUDE ALL FAILURES; PSNR DIFFERENCES USE DIRECT SOURCE-PAIRED 95% INTERVALS. ALL FIVE METRICS AND PAIRED COMPARISONS ARE IN SUPPLEMENT S2.

SNR / method	PSNR↑	DISTS↓	DreamSim↓
4 / WHOLE	15.11	0.2580	0.3247
4 / NeST-Com	15.61	0.2363	0.2740
4 / Ada-TokenCom	11.32	0.2996	0.3847
NeST–WHOLE ΔPSNR: +0.504 [0.440, 0.573] dB			
10 / WHOLE	16.82	0.2030	0.2049
10 / NeST-Com	17.41	0.1922	0.1837
10 / Ada-TokenCom	13.16	0.2367	0.2582
NeST–WHOLE ΔPSNR: +0.592 [0.544, 0.643] dB			
19 / WHOLE	18.64	0.1718	0.1458
19 / NeST-Com	19.63	0.1554	0.1150
19 / Ada-TokenCom	17.69	0.1496	0.1078
NeST–WHOLE ΔPSNR: +0.994 [0.940, 1.048] dB			

15.75/14.95/13.50% at 9 dB, 0.10–0.20% at 9.5 dB and zero observed at 10 dB. Overall quality includes gray failures (S5).

IV. CONCLUSION

NeST-Com organizes next-scale source coding, channel protection and prefix-constrained reconstruction within a fixed-budget digital communication framework. Its actual-length within-scale fitting retains scale-wise probability evaluation and converts otherwise unused frame capacity into additional source information. Matched controls and receiver interventions explain

TABLE IV

FROZEN POLICIES: MCS (q, r) (BITS/SYMBOL, CODE RATE) AND TARGET CEILING e, WITH FEASIBLE FALLBACK. ADA'S CEILING IS 256 TOKENS. BUDGETS ARE CALIBRATED INDEPENDENTLY.

N/SNR	a_{W}	e_{W}	a_{N}	e_{N}	a_{Ada}
1024/4	$2, 2/3$	$(7, 0)$	$2, 2/3$	$(7, 25)$	$2, 2/3$
1024/7	$4, 1/2$	$(8, 0)$	$4, 1/2$	$(8, 0)$	$4, 1/2$
1024/10	$4, 2/3$	$(9, 0)$	$4, 2/3$	$(8, 126)$	$4, 2/3$
1024/19	$6, 5/6$	$(10, 0)$	$6, 5/6$	$(10, 0)$	$4, 5/6$
512/4	$2, 2/3$	$(6, 0)$	$2, 2/3$	$(5, 27)$	$2, 2/3$
512/10	$4, 2/3$	$(7, 0)$	$4, 2/3$	$(7, 0)$	$4, 2/3$
512/19	$6, 5/6$	$(8, 0)$	$6, 5/6$	$(8, 84)$	$6, 5/6$

the reconstruction gains; complete-system results characterize performance across budgets and the associated quality–cost tradeoffs.

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